

Marzia Tahsin

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EDUCATION

University of Maryland, Baltimore County (UMBC)

Graduation date: Dec 2026

B.S. Computer Science | GPA 3.56 | CWIT Affiliate

SKILLS

Programming Languages: C++, C, Python, JavaScript, HTML/CSS, SQL, Qt/QML

Libraries & Frameworks: Pandas, NumPy, Matplotlib, scikit-learn, PyQt6

Technical Skills: Linux, Git/GitHub, Firebase, React, .NET, SQLite

Certifications (Coursera): AWS Fundamentals, Foundations of Cybersecurity by Google

EXPERIENCE

Head Teaching Fellow | University of Maryland Baltimore County (UMBC) | MD

September 2024 - Present

- Lead weekly discussion sessions and hold office hours, supporting students in undergraduate coding coursework
- Coordinate course logistics, manage TA scheduling, and ensure grading consistency across a team of TAs
- Mentor and onboard new TAs, fostering a cohesive and high-quality instructional team

Research Assistant | UMBC Department of Computer Science, Baltimore, MD

September 2025 - Present

- Contributing to research on an AI facilitator for collaborative visual sensemaking and human-AI interaction
- Analyze participant dialogue and code interaction data using the NOVIS framework
- Build prompt libraries and discourse modules for an autonomous AI agent; collaborate with a multidisciplinary team on data visualization

AI4ALL Student Fellow | AI4ALL Ignite Program

May 2025 - September 2025

- Competitively selected for a program focused on AI fundamentals, algorithmic fairness, and ethical AI applications
- Completed workshops and mentorship sessions on core machine learning concepts and responsible AI design
- Built a capstone project on EEG-based emotion recognition using Random Forest Classifier to classify emotional states from brainwave data

PROJECTS

ASC Check-In System (UMBC Capstone)

- Built a full-stack check-in app using React, .NET REST API, Firebase (Firestore + auth), and Vercel; delivered SRS, UI design, and testing documentation
- Implemented real-time student check-in/out, tutor scheduling, and an admin dashboard for session tracking and reporting

EEG-Based Emotion Recognition (AI4ALL Capstone)

- Trained a Random Forest Classifier on EEG brainwave data to classify emotional states, achieving 62.9% accuracy despite class imbalance and overlapping neural patterns
- Feature importance analysis revealed distributed emotion representation across multiple brain regions; presented findings via research poster and live demo at AI4ALL Final Presentation

MyItinerary (Itinerary app)

- Built a Python/PyQt6 desktop app with SQLite for data management, Leaflet.js for interactive mapping, and budget tracking
- Integrated an Ollama AI chatbot for personalized travel recommendations

The Sound of Globalization - Spotify Global Music Analysis

- Analyzing Spotify chart data across countries to identify shared audio features in global hits using unsupervised clustering and a custom "global presence score" (streams, chart weeks, country reach)
- Building an interactive Python dashboard with globe map, scatter plots, box plots, and cluster visualizations across 19 audio attributes

Linux Kernel Priority Queue

- Implemented four custom Linux kernel system calls to manage a thread-safe priority queue (init, free, enqueue, dequeue) with mutex locks for concurrent access
- Supports three priority levels with unique node IDs; validated with a proof-of-concept program and automated concurrency stress tests

SimpleFS — FUSE-Based Filesystem

- Built a custom FUSE filesystem in C that stores files inside a disk image using a fixed superblock and file table format
- Supports core filesystem operations (create, read, write, delete) entirely in userspace